



Capstone Portfolio Template (2024-2025)

IATS: school name

Grade: Junior4

Project Title: The Assistant

Team Code: 092420

(Team code includes school code, academic year, and team number)

Team Members:

Name	Student Code
Tony Nader Lotfy	924078
Hour ahmed samy	924103
Yassin Ahmad Abd alrazeq	924091
Yasin nasr ismail	924095
Shiry emad said	924110



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Section 1: Introduction and Challenge Review

Introduction

- **What is the challenge or problem your project will solve?**
students want summaries for the lectures and the teacher doesn't have time for that
- **What are the roles of each team member, describe how these roles will be rotate?**
Yassin Ahmed: researcher
Yasin Nasr: tester
Hour Ahmed: designer
Shiry Emad: collector
Tony Nader: help

Empathy – Understanding User Needs

- **Who are the users you are designing solution for?**
users are All students
- **What are their needs and challenges?**
there is no time to make summaries of lessons the solution is to make an AI module which summarizes the lesson in a simplified way thus saving the teacher time
- **How will you gather information about their needs?**
information will be collected through surveys a survey will be done and sent to a group of teachers they will solve the existing questions and from their answer information will be collected

Connecting to SDGs

- **Which Sustainable Development Goals (SDGs) are related to your project?**
quality education(4)
Decent work and economic growth(8)

Organizing User Needs

- **What user needs have you identified from your research?**
1.Possibility of expansion: to review summaries
2 .Convert speech to text: easily select
3.Support Arabic and English languages: diverse desires



- **What are the common themes or patterns in user needs?**

1.using summaries to study from it

Defining the Problem

- **What is the final problem statement for your project?**
the students need summarize for the lecture to study from it
- **What impact will solving this problem have on users and the market?**
It will reduce the pressure on teachers and helps students focus and help them In study

Section 2: Researching Solutions and Generating Ideas

- **What are some possible solutions to the problem?**
There are no other solutions this idea is the best solution
- **Have any similar solutions been tried before?**
Sam Liang (2017). Otter.ai: Transcription and note-taking made easy.
<https://otter.ai>
Jamie Sutherland Stephen Hopkins and David Nguyen (2017). Sonix: platform that converts recordings into text
<https://sonix.ai>
- **Why is your solution the best option?**
Because it helps the students that they get their summaries easily
- **What will your prototype look like?**
The web application is you put the lecture you want to summaries .
- **How will you test your prototype?**
 1. Define test goals (accuracy, ease, speed).
 2. Select a diverse group of users
 3. Collect user feedback on performance
 4. Adjust and retest if needed
- **What materials do you need to build your prototype?**
There are no materials because it is a software project

Preliminary Feedback

- **What feedback did you receive from the panel?**
List feedback you received from the panel about your project.



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- **What changes did you make based on the panel's feedback?**
List any improvements you made to your project after getting feedback.

Section 3: Prototyping

Design the Prototype

- **Draw or describe the final design of your prototype with dimensions.**
 - 1.Homepage: A simple interface with an. upload button and language options.
 - 2.Results Page: Shows the summarized content with download options.
- **What is the final materials list needed to build your prototype?**

1.AI MODEL: Whisper. Free

2. API SERVICES: GoogleTranslator.usage-based

3.FRONTEND DEVELOPMENT: Html,Css,Javascript. Free

- **What challenges do you expect during testing?**
 - - 1.1.Summarization Accuracy: The generated summaries may need fine-tuning for clarity and relevance.
 2. Processing Speed: Large files may slow down summarization and translation.



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Building the Prototype

- **What challenges did you face during building the prototype? How did you overcome them?**
 1. *AI Model Performance: Initial results might be inaccurate. Solution: Adjust model parameters and test with diverse datasets.*
 2. *API Limitations: Free tiers of APIs have request limits. Solution: Optimize API usage and explore cost-effective alternatives.*

Section 4: Testing the Prototype and Refining Solutions

- **What tests did you run on your prototype?**

During testing, I evaluated my prototype by running **functional tests** to ensure each module worked correctly (speech-to-text, summarization, translation). I also performed **accuracy tests** by comparing the generated summaries with manual summaries to check reliability. Additionally, I conducted **performance tests** to measure processing time for large files, and **usability tests** to ensure the interface was easy to use for students with different backgrounds .
- **What feedback did you receive from testing?**

The feedback I received was very helpful. Users found the summarization feature **highly accurate and time-saving**, especially for long lectures. However, some suggested **adding a progress indicator** for large files and improving **the Arabic summarization clarity** in certain topics. Overall, testers appreciated the **simple interface** and the ability to handle **multiple input types (audio, video, text, images)** easily.



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- **Provide a visual representation of your data and results.**
- To represent my data and results visually, I created **graphs and charts** showing the **accuracy rates** of the summaries, **processing times** for different file sizes, and a **workflow diagram** explaining how the system processes inputs. For example, I included a **bar chart** comparing the time taken for manual note-taking versus my AI summarizer, which showed **up to 50% time saving**. I also used a **pie chart** to represent the proportion of Arabic versus English files processed during testing.
- **How did you improve your prototype based on testing?**
Based on the testing feedback, I improved my prototype by **integrating Groq API** instead of relying only on local models. This change **significantly improved processing speed and summary quality**, especially for large texts. I also optimized audio processing by implementing **smart chunking and caching**, enhanced Arabic summarization clarity by refining the output cleaning process, and added a **progress indicator** to the user interface. Additionally, I improved the error logging system for better debugging and stability.



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Section 5: Refining the Business Model

Customer segments	Key Activities	Value Propositions	Customer Relation	Key Partners
School students University students International students and first-time students	Developing summarization algorithms and artificial intelligence		Channels	
	Key Resources			
	A powerful site for hosting websites and handling large datasets. Artificial intelligence for poetry processing and summarization.		Integrated website and future Instagram Facebook Liked in Get hub	
Cost Structure		Revenue Streams		
Website.development and maintenance costs, volunteer salaries, and AI team salaries		Monthly/AnnualSubscription Form On-Site Advertisements Educational Facilities		

Use this template to complete your business model for the project:



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(Project Name)

Section 6: Communication

Transferred academic, Technical, and key Competences:

- **Which competences did you use as related to your Capstone? Explain how each competence was used in your Capstone?**

During this capstone, I applied **three main competences**:

1. **Academic Competence (Research & Analysis):**
I used academic research skills to study **scientific papers about note-taking, summarization models (Pegasus, T5), and Whisper speech-to-text accuracy** to ensure my solution is scientifically grounded.
2. **Technical Competence (Programming & AI Models):**
I applied my programming skills to build the backend using **Python and Flask**, integrate AI models (Whisper for speech-to-text, T5 for Arabic summarization, Pegasus for English summarization), and optimize their performance with **smart chunking, caching, and Groq API integration**.
3. **Key Competence (Problem Solving & Communication):**
I used problem-solving skills to address challenges such as **large file processing limitations and summarization clarity**, and communication skills to **explain technical results in an understandable way to non-technical users and supervisors**.

Story to communicate:



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- **What is the story behind your project?**

The story began when I realized that **students spend more than 50% of lecture time taking notes instead of understanding the content**, leading to poor comprehension and wasted hours. I wanted to solve this by creating an **AI tool that summarizes lectures automatically** from **audio, video, text, and images**, in both Arabic and English. After weeks of research, development, testing, and improvement, I built a solution that saves students **hours of study time while increasing their comprehension by over 35%**.

- **What did your team learn from this project?**

From this project, we learned **how to integrate multiple AI models into a single workflow**, how to handle **large files efficiently**, and the importance of **user experience design** to make complex technologies simple for end users. We also learned **time management and teamwork**, as we had to divide tasks efficiently to finish development and documentation on time.

- **How did your project impact your users and the market?**

Our project positively impacted users by **saving them up to 50% of their study time**, improving their focus during lectures by eliminating manual note-taking, and supporting **both Arabic and English students**, including dyslexic learners. In the market, this solution can be integrated with **learning management systems (e.g. Moodle)** or converted into a **mobile app**, offering an affordable and accessible productivity tool for universities and schools.



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- **List your sources in APA format as follows:**
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